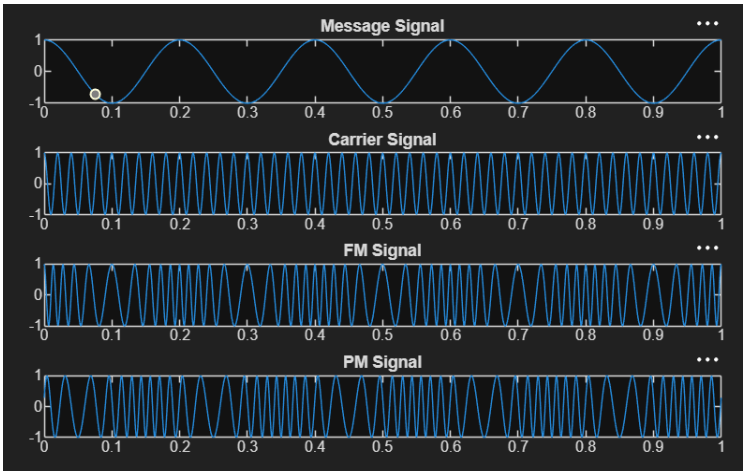
GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) AND OF PHASE MODULATED (PM) SIGNALS USING MATLAB

```
clc;
clear;
close all;
Am=1; Ac=1; fm=5; fc=50; beta=5; kp=5;
fs=1000; t=0:1/fs:1; m =
Am*cos(2*pi*fm*t); % Message signal


c = Ac*cos(2*pi*fc*t); % Carrier signal fm_sig =
Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM pm_sig
= Ac*cos(2*pi*fc*t + kp*m); % PM Df = beta*fm; %
Frequency deviation fprintf('Frequency Deviation = %d
Hz\n',Df); fprintf('FM Modulation Index = %d\n',beta);
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
subplot(4,1,4), plot(t,pm_sig), title('PM Signal')
```